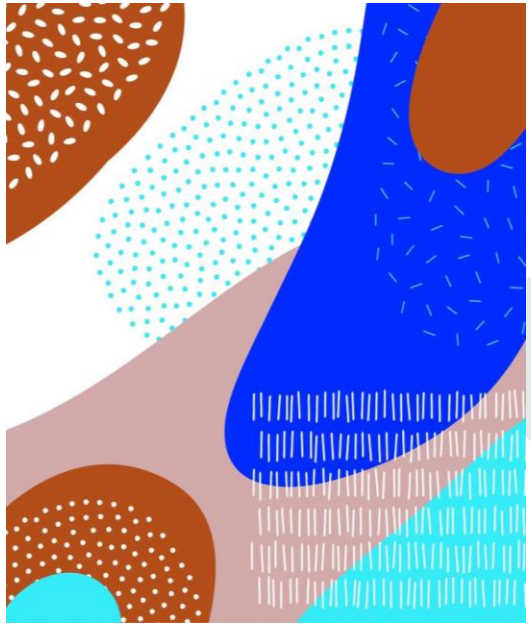

Introduction to Problem Solving

Meaning of a Problem

A **problem** is a situation where:

We have a **goal**

We do **not know immediately** how to achieve that goal

in simple words:

A problem exists when there is a **difference between what we have and what we want**.

Simple Examples

You want to score good marks, but you don't know how to prepare

A shop owner wants more customers, but sales are low

A computer program gives the wrong output

All these are **problems** because the solution is not obvious.

What is Problem Solving?

Problem solving is the process of:

- Thinking logically
- Finding a clear and correct solution
- Applying that solution step by step

Problem solving is **not guessing** and **not memorization**.
It is **logical thinking**.

Problem solving = Thinking + Planning + Execution

Why Problem Solving is Important

3.1 In Daily Life

- Choosing the best option
- Managing time
- Handling unexpected situations

3.2 In Education

- Understanding subjects instead of memorizing
- Solving numerical and logical questions
- Writing correct programs

3.3 In Career and Industry

- Employers prefer problem solvers
- Every job requires decision-making
- AI, Data Science, and Programming are based on problem solving

Problem Solving as a Skill

Problem solving is a **skill**, not a talent.

Skills can be learned	Skills improve with practice

Anyone can become a good problem solver by:

Practicing logical thinking	Solving more problems	Learning from mistakes

Characteristics of a Good Problem Solver

A good problem solver:

- Understands the problem clearly
- Thinks step by step
- Breaks big problems into small parts
- Tries different solutions if needed
- Learns from failure

Good problem solvers are **patient and logical**.

Types of Problems

6.1 Well-Defined Problems

- Clear input
- Clear output
- Clear method

Example:
Find the sum of two numbers.

6.2 Ill-Defined Problems

- No clear solution
- Multiple possible answers

Example:
How to become successful in life.

6.3 Simple Problems

- Few steps
- Easy to solve

Example:
Check whether a number is even or odd.

6.4 Complex Problems

- Many steps
- Need deep thinking

Example:
Designing a university management system.

Steps in Problem Solving

Every problem can be solved using the following **five basic steps**.

Step 1: Understand the Problem

Ask questions:

- What is the problem asking?
- What information is given?
- What is required as output?

Example: If a question asks to calculate average marks, understand:

- How many subjects?
- Total marks?
- Formula?

Analyze the Problem

Break the problem into smaller parts.

- Identify inputs
- Identify outputs
- Identify conditions

This makes the problem easier.

Design the Solution

Plan the solution before doing it.

This can be done using:

- Algorithms
- Flowcharts
- Pseudocode

Designing saves time and avoids mistakes.

Step 4: Implement the Solution

Write the program

Follow the designed steps

Computer will follow instructions exactly.

Step 5: Test and Improve

- Check output
- Find errors
- Improve logic

Testing ensures correctness.

Simple Daily Life Example

Problem:
You are getting late for college.

Solution using problem-solving steps:

- Understand: Time is short
- Analyze: Transport options, distance
- Design: Leave early, choose faster route
- Implement: Follow plan
- Test: Reach on time or improve next day

Same logic applies to **programming problems**.

Problem Solving in Programming

Computers:

- Do not think
- Do not understand language
- Only follow instructions

So the programmer must:

- Think clearly
- Give step-by-step instructions

That is why:

Good programmers are good problem solvers.

Common Mistakes in Problem Solving

- Starting solution without understanding problem
- Ignoring conditions
- Not checking output
- Making solution too complex
- Avoid these mistakes by **planning first**.

Final Thought

If you learn how to think, you can learn anything.”